MILITARY SPECIFICATION

RESISTORS, VARIABLE, WIRE-WOUND, PRECISION, STYLE RR3800

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for style RR3800, variable, wire-wound, precision resistors.

1.2 Classification.

1.2.1 Type designation. The type designation shall be in the following form:

Single-section (cup) resistors

RR3800	A	2	A	7	J	101
Style (1.2.1.1)	Function and shaft length (1.2.1.2)	Class (1.2.1.3)	Resistance temperature characteristic (1.2.1.4)	Rotational life characteristic (1.2.1.5)	Function conformity tolerance and resistance tolerance characteristic (1.2.1.6)	Resistance (1.2.1.7)

1.2.1.1 Style RR3800. Style RR3800 is applicable to this specification.

1.2.1.2 Function and shaft length. The function and shaft length applicable to this specification shall be identified by a single symbol in accordance with table I.

TABLE I. Function and shaft length.

Symbol	Shaft length Tolerance ±1/32 (.79)	Function
A - - - - -	3/8 (9.53)	Linear
B - - - - -	1/2 (12.70)	Linear
C - - - - -	5/8 (15.88)	Linear
D - - - - -	3/4 (19.05)	Linear
E - - - - -	7/8 (22.23)	Linear
F - - - - -	1 (25.40)	Linear

1.2.1.3 Class. The class applicable to this specification shall be symbols 2 and 3.

1.2.1.4 Resistance temperature characteristic. The resistance temperature characteristics applicable to this specification are symbols A to H, inclusive.

1.2.1.5 Rotational life characteristics. The rotational life characteristics applicable to this specification are symbols 7, 8, and 9. This potentiometer is capable of operation up to 100 RPM.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:49Z
Check the source to verify that this is the current version before use.

1.2.1.6 Function conformity tolerance and resistance tolerance characteristics. The function conformity tolerance and resistance tolerance characteristics applicable to this specification are symbols G to L, inclusive and S to X, inclusive.

1.2.1.7 Resistance. The nominal total resistance values applicable to this specification shall be in accordance with table II.

TABLE II. Nominal total resistance and nominal resolution characteristics.

Nominal total resistance value	Nominal resolution
(Ohms)	(Percent)
100	.075
200	.061
500	.048
1,000	.038
2,000	.035
5,000	.028
10,000	.031
20,000	.024
50,000	.018
100,000	.014

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent herein:

SPECIFICATION

MILITARY

MIL-R-12934 - Resistors, Variable, Wire-wound, Precision.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-12934 and as specified her herein.

3.2 Design and construction. The resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Minimum resistance. Minimum total resistance value shall be 100 ohms.

3.4 Maximum resistance. Maximum total resistance value shall be 100,000 ohms.

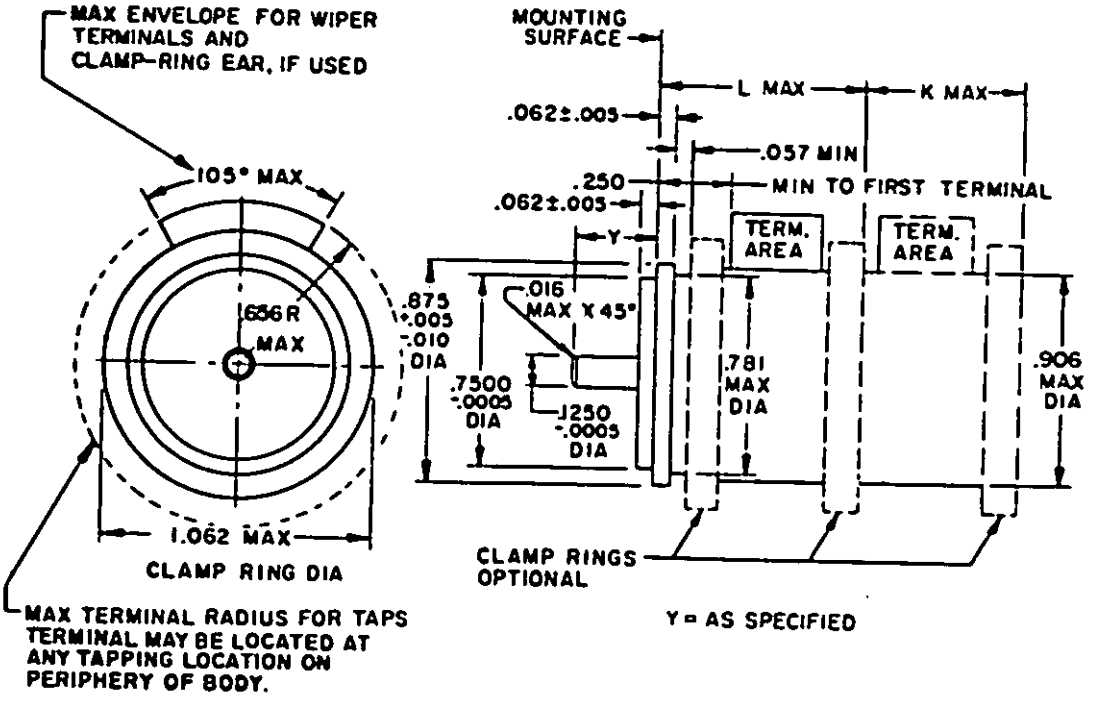
3.5 Mechanical travel. Mechanical travel shall be $1,800\ ^{+10^\circ}_{-0^\circ}$.

3.6 Actual electrical travel. The actual electrical travel shall be $1,800\ ^{+4^\circ}_{-0^\circ}$.

3.7 Ganged cups. There shall be no more than 4 cups ganged.

3.8 Phasing. For independent linearity the phasing between the cups shall be within $\pm 1^\circ$. This shall be measured, with respect to the first cup, starting from the counterclockwise end point.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:49Z
Check the source to verify that this is the current version before use.



INCHES	MM	INCHES	MM
.0005	.01	.250	6.35
.005	.13	.656	16.66
.010	.25	.7500	19.05
.016	.41	.781	19.84
.057	1.45	.875	22.23
.062	1.57	.906	23.01
.1250	3.18	1.062	26.97

Style	Turns	L (max)	K (max)	Maximum continuous working voltage (volts)	Power rating (watts) 1/	Maximum starting and running torque (ounce-inch)				Stop-torque (pound-inch)
						Five-turn single cup		Per each additional five-turn cup		
						starting	running	starting	running	
RR3800	5	1.219 (30.96)	.968 (24.59)	500	1.5	.5	.4	.4	.3	3

1/ When 5 turn units are ganged all cups will be 75% of rated wattage.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Millimeters are in parentheses.

FIGURE 1. Style RR3800.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:49Z
Check the source to verify that this is the current version before use.

3.9 Function. The type of function applicable to this specification is linear function (independent linearity).

3.10 Weight. The maximum weight shall be .06 pound (27.22 grams), and for each additional cup .04 pound (18.14 grams).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection shall be in accordance with MIL-R-12934 and as specified herein.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-12934.

6. NOTES

6.1 The notes specified in MIL-R-12934 are applicable to this specification.

6.2 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - MI, MU, EL
Navy - EC, AS, OS
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - ME, AT, AV
Navy -
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5905-0790-3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:49Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Communications - Electronics Command
ATTN: DRSEL-ED-TO
Fort Monmouth, NJ 07703

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:49Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:49Z
Check the source to verify that this is the current version before use.